Chart of Accounts and System Rules

This document explains how the accounting system is structured and all the business rules enforced in the LUBOSS 95 Village Banking system.

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Chart of Accounts

The system uses a double-entry bookkeeping approach with a hierarchical chart of accounts. Accounts are organized by type and can be either organization-level (shared) or member-specific (individual).

Account Types

The system uses five standard accounting account types:

1. **ASSET** - Resources owned by the organization or receivables
2. **LIABILITY** - Obligations owed by the organization
3. **INCOME** - Revenue earned by the organization
4. **EXPENSE** - Costs incurred by the organization
5. **EQUITY** - Net worth of the organization

Organization-Level Accounts (Core Accounts)

These accounts are created during initial system setup and are shared across all members:

Asset Accounts

- `BANK_CASH` - Main bank account where all cash deposits are recorded
- `SOC_FUND_REC` - Social Fund Receivable (organization's receivable from all members for social fund contributions)
- `ADM_FUND_REC` - Admin Fund Receivable (organization's receivable from all members for admin fund contributions)

Liability Accounts

- `SOCIAL_FUND` - Social Fund Payable (legacy account, maintained for compatibility)
- `ADMIN_FUND` - Admin Fund Payable (legacy account, maintained for compatibility)

Income Accounts

- `INTEREST_INCOME` - Interest income earned from loans
- `PENALTY_INCOME` - Penalty income from late payments and compliance violations

Equity Accounts

- `CARRY_FORWARD` - Carry-forward reserve for end-of-cycle distributions

Member-Specific Accounts

These accounts are created automatically for each member when needed. Each member has their own set of accounts:

Member Savings Account

- **Account Code**: `MEM_SAV_{member_short_id}`
- **Account Type**: LIABILITY
- **Purpose**: Tracks member's accumulated savings
- **When Created**: Automatically created when member makes their first deposit or when running `setup_ledger_accounts.py`

Member Social Fund Account

- **Account Code**: `MEM_SOC_{member_short_id}`
- **Account Type**: ASSET (for receivable tracking)
- **Purpose**: Tracks member's social fund contributions and outstanding balance
- **When Created**:
  - Automatically created when member makes their first declaration for a cycle
  - Or when running `setup_ledger_accounts.py`

Member Admin Fund Account

- **Account Code**: `MEM_ADM_{member_short_id}`
- **Account Type**: ASSET (for receivable tracking)
- **Purpose**: Tracks member's admin fund contributions and outstanding balance
- **When Created**:
  - Automatically created when member makes their first declaration for a cycle
  - Or when running `setup_ledger_accounts.py`

Member Penalties Payable Account

- **Account Code**: `PEN_PAY_{member_short_id}`
- **Account Type**: LIABILITY
- **Purpose**: Tracks penalties owed by the member
- **When Created**:
  - Automatically created when a penalty is approved for the member
  - Or when running `setup_ledger_accounts.py`

Member Loans Receivable Account

- **Account Code**: `LOAN_REC_{member_short_id}`
- **Account Type**: ASSET

- **Purpose**: Tracks loans issued to the member (principal amount)
- **When Created**:
  - Automatically created when a loan is approved and disbursed
  - Or when running `setup_ledger_accounts.py`

Account Naming Convention

- **Organization accounts**: Use descriptive codes (e.g., `BANK_CASH`, `INTEREST_INCOME`)
- **Member accounts**: Use prefix + member short ID pattern:
  - `MEM_SAV_` - Member Savings
  - `MEM_SOC_` - Member Social Fund
  - `MEM_ADM_` - Member Admin Fund
  - `PEN_PAY_` - Penalties Payable
  - `LOAN_REC_` - Loans Receivable

The `member_short_id` is the first 8 characters of the member's UUID (without hyphens).

Account Creation Process

Initial Setup

When the system is first set up, run the following script to create all core accounts:

```
python scripts/setup_ledger_accounts.py
```

This script:

1. Creates all organization-level accounts if they don't exist
2. Creates member-specific accounts for all existing members
3. Prevents duplicate account creation

Dynamic Account Creation During Operation

The system automatically creates member accounts as needed during normal operations:

1. First Declaration for a Cycle

When a member makes their **first declaration** for a cycle:

- **Social Fund Account** is created (if it doesn't exist)
- **Admin Fund Account** is created (if it doesn't exist)
- **Organization receivable accounts** (`SOC_FUND_REC`, `ADM_FUND_REC`) are created if missing
- **Initial required amounts** are posted:
  - Debit: Member's Social Fund account (annual required amount)
  - Credit: Organization's `SOC_FUND_REC` account
  - Debit: Member's Admin Fund account (annual required amount)
  - Credit: Organization's `ADM_FUND_REC` account

2. Deposit Approval

When the Treasurer approves a deposit:

- **Member Savings Account** is created if missing
- **Member Social Fund Account** is created if missing (for social fund payments)
- **Member Admin Fund Account** is created if missing (for admin fund payments)
- **Member Penalties Payable Account** is created if missing (for penalty payments)
- **Organization receivable accounts** are created if missing

3. Loan Disbursement

When a loan is approved and disbursed:

- **Member Loans Receivable Account** is created if missing
- Journal entries are posted to record the loan

4. Penalty Approval

When the Treasurer approves a penalty:

- **Member Savings Account** is created if missing (penalties are deducted from savings)
- **Member Penalties Payable Account** is created if missing
- Journal entries are posted to record the penalty

Account Lifecycle

- **Creation**: Accounts are created on-demand when first needed
- **Status**: All accounts are marked as `is_active = True` by default
- **Deactivation**: Accounts can be deactivated but are never deleted (for audit trail)
- **Uniqueness**: Account codes must be unique across the system

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System Rules and Validations

The system enforces numerous business rules to ensure data integrity and compliance with village banking policies. These rules are categorized below:

1. Declaration Rules

Declaration Creation

- ✅ **One Declaration Per Month**: A member cannot create multiple declarations for the same month and cycle
- ✅ **Active Member Required**: Only members with `ACTIVE` status can create declarations
- ✅ **Active Cycle Required**: Declarations can only be created for active cycles

Declaration Editing

- ✅ **Current Month Only**: Declarations can only be edited for the current month (cannot edit previous months)
- ✅ **Status Restriction**: Only `PENDING` declarations can be edited (unless explicitly allowed after rejection)
- ✅ **No 20th Day Restriction**: Current month declarations can be edited anytime (previous restriction removed)
- ✅ **Penalty Inclusion**: Only `APPROVED` penalties are included in declaration penalty lists (not `PENDING` or `PAID`)
- ✅ **Cycle-Defined Penalty Filter**: Cycle-defined penalties are only included if their phase's `auto_apply_penalty` flag is enabled

First Declaration Special Handling

- ✅ **Initial Posting**: On the first declaration for a cycle, the system automatically posts:
  - Annual Social Fund required amount as a **debit** to member's Social Fund account
  - Annual Admin Fund required amount as a **debit** to member's Admin Fund account
  - Corresponding **credits** to organization receivable accounts
- ✅ **Duplicate Prevention**: Initial postings are only made once per cycle (checked before posting)

2. Deposit Approval Rules

Deposit Proof Validation

- ✅ **Status Check**: Both `SUBMITTED` and `REJECTED` deposit proofs can be approved (rejected proofs can be approved after member response)
- ✅ **Declaration Required**: Deposit proof must be linked to a valid declaration
- ✅ **Amount Validation**: Deposit amount must match declaration total (with small tolerance for rounding: ±0.01)
- ✅ **Status Update**: On approval, deposit proof status changes to `APPROVED` and declaration status changes to `APPROVED`

Late Deposit Period

- ✅ **Deposit Window**: Deposits for a declaration's effective month are accepted from:
  - **Start**: `monthly_start_day` (typically 26th) of the effective month
  - **End**: `monthly_end_day` (typically 5th) of the **next** month
  - Example: For January 2026 declaration, deposit period is January 26, 2026 to February 5, 2026
- ✅ **Late Penalty**: Deposits submitted after the end date trigger automatic late deposit penalties (if `auto_apply_penalty` is enabled for the deposits phase)

Journal Entry Posting

When a deposit is approved, the system creates a balanced journal entry:

Base Entry:

- Debit: `BANK_CASH` (full deposit amount - cash received)

For Savings:

- Credit: Member Savings Account (declared savings amount)

For Social Fund Payment:

- Credit: Member Social Fund Account (payment amount - reduces balance due)
- Note: Required amount was posted as debit on first declaration; payment credit reduces balance
- Balance calculation: Debits (required) - Credits (payments) = Balance Due

For Admin Fund Payment:

- Credit: Member Admin Fund Account (payment amount - reduces balance due)
- Note: Required amount was posted as debit on first declaration; payment credit reduces balance
- Balance calculation: Debits (required) - Credits (payments) = Balance Due

For Penalties:

- Credit: Member Penalties Payable Account (reduces liability)
- **Penalty Status Update**: When penalties are paid via deposit approval, matching `APPROVED` penalty records are marked as `PAID`

For Interest on Loan:

- Credit: `INTEREST_INCOME` (income earned)

For Loan Repayment:

- Credit: Member Loans Receivable Account (reduces asset/loan balance)

- ✅ **Balance Validation**: All journal entries must balance (total debits = total credits)

- ✅ **Account Creation**: Missing member accounts are automatically created during approval

3. Loan Rules

Loan Application

- ✅ **Active Member**: Only active members can apply for loans
- ✅ **Active Cycle**: Loans can only be applied for in active cycles
- ✅ **Eligibility Check**: Loan amount must not exceed borrowing limit based on:
  - Member's savings balance
  - Credit rating tier multiplier
  - Cycle-specific borrowing limit policy
- ✅ **Single Active Loan**: Members can only have one active loan at a time
- ✅ **Pending Application Edit**: Pending loan applications can be edited or withdrawn

Loan Approval and Disbursement

- ✅ **Status Check**: Only `PENDING` loan applications can be approved
- ✅ **Approval Required**: Loan must be approved before disbursement
- ✅ **Journal Entry**: On disbursement:

    - Debit: Member Loans Receivable Account
    - Credit: `BANK_CASH`
- ✅ **Account Creation**: Member Loans Receivable account is created if missing

Loan Repayment

- ✅ **Declaration-Based**: Loan repayments are declared in monthly declarations
- ✅ **Principal vs Interest**: Repayments are split into principal and interest based on loan terms
- ✅ **Balance Reduction**: Principal portion reduces the loan receivable balance

4. Penalty Rules

Penalty Status Workflow

Penalties follow a three-stage status workflow:

1. `PENDING`: Penalty created by compliance officer, awaiting treasurer approval
2. `APPROVED`: Penalty approved by treasurer, posted to ledger, included in declarations
3. `PAID`: Penalty paid via deposit approval, excluded from future declarations

Penalty Creation

- ✅ **Penalty Type Required**: Penalty record must have a valid penalty type
- ✅ **Fee Amount Validation**: Penalty type must have a valid fee amount (> 0)
- ✅ **Cycle-Defined Penalties**: System automatically creates penalties for:
    - **Late Declaration**: Created when declaration is made after `monthly_end_day` of effective month (if `auto_apply_penalty` enabled)
    - **Late Deposits**: Created when deposit is submitted after `monthly_end_day` of next month (if `auto_apply_penalty` enabled)
    - **Late Loan Application**: Created when loan application is submitted after `monthly_end_day` (if `auto_apply_penalty` enabled)
- ✅ **Auto-Approval**: Cycle-defined penalties are created with `APPROVED` status (no treasurer approval needed)
- ✅ **Duplicate Prevention**: System checks for existing penalties by member, penalty type, and effective month/date before creating

Penalty Approval (Manual)

- ✅ **Status Check**: Only `PENDING` penalties can be approved
- ✅ **Account Creation**: Member Savings and Penalties Payable accounts are created if missing
- ✅ **Journal Entry**: On approval:
    - Debit: Member Savings Account (penalty charged to member)
    - Credit: Member Penalties Payable Account (increases liability)
    - Credit: `PENALTY_INCOME` (income earned)
- ✅ **Status Update**: Penalty status changes from `PENDING` to `APPROVED`

Penalty Payment

- ✅ **Payment via Deposit**: Penalties are paid when member includes penalty amount in declaration and deposit is approved
- ✅ **Status Update**: When deposit is approved, matching `APPROVED` penalties are marked as `PAID` (oldest first)
- ✅ **Journal Entry**: Payment is recorded as:
  - Credit: Member Penalties Payable Account (reduces liability)
- ✅ **Exclusion**: `PAID` penalties are excluded from:
  - Future declaration penalty lists
  - Transaction history PenaltyRecord section (shown only as journal entry lines)

5. Accounting Rules

Journal Entry Validation

- ✅ **Double-Entry Balance**: Every journal entry must balance (total debits = total credits)
- ✅ **Account Existence**: All accounts referenced in journal entries must exist
- ✅ **Account Type Consistency**: Debits and credits must align with account types:
  - **ASSET accounts**: Debits increase, Credits decrease
  - **LIABILITY accounts**: Credits increase, Debits decrease
  - **INCOME accounts**: Credits increase (income), Debits decrease
  - **EXPENSE accounts**: Debits increase (expenses), Credits decrease

Balance Calculations

Savings Balance:

- Sum of all credits to member savings account (deposits)
- Minus any debits (penalties, withdrawals)

Loan Balance:

- Sum of all debits to member loans receivable account (loan disbursements)
- Minus all credits (loan repayments)

Social Fund Balance:

- Shows accumulated payments made (sum of payment debits/credits)
- Initial required amount is posted as debit on first declaration
- Payments are posted as debits to member account

Admin Fund Balance:

- Shows accumulated payments made (sum of payment debits/credits)
- Initial required amount is posted as debit on first declaration
- Payments are posted as debits to member account

Penalties Balance:

- Sum of all credits to penalties payable account (penalties assessed)
- Minus all debits (penalties paid)

- Shows outstanding penalties owed by member

Penalty Status Tracking:

- **PENDING**: Awaiting treasurer approval, not included in declarations
- **APPROVED**: Approved and posted, included in declarations, shown in transaction history
- **PAID**: Paid via deposit, excluded from declarations, shown only as payment journal entries in transaction history

6. Member Status Rules

Member Activation

- ✅ **Approval Required**: Members must be approved by Chairman before activation
- ✅ **Status Transition**: Members move from `PENDING` → `ACTIVE` upon approval
- ✅ **Suspension**: Active members can be suspended (status: `SUSPENDED`)
- ✅ **Reactivation**: Suspended members can be reactivated by Chairman

Member Access

- ✅ **Active Status Required**: Only `ACTIVE` members can:
  - Create declarations
  - Apply for loans
  - Upload deposit proofs
  - Access member dashboard features
- ✅ **Suspended Members**: Suspended members cannot perform transactions but can view their account history

7. Cycle Management Rules

Cycle Creation

- ✅ **Unique Year**: Each cycle must have a unique year
- ✅ **Date Validation**: Start date must be before end date (if end date is set)
- ✅ **Required Amounts**: Social Fund and Admin Fund required amounts can be set per cycle

Cycle Activation

- ✅ **Single Active Cycle**: Only one cycle can be active at a time
- ✅ **Status Transition**: Cycles move from `DRAFT` → `ACTIVE` upon activation

Cycle Closure

- ✅ **Active Status Required**: Only `ACTIVE` cycles can be closed
- ✅ **Status Transition**: Cycles move from `ACTIVE` → `CLOSED` upon closure
- ✅ **Reopening Restriction**: Only closed cycles can be reopened
- ✅ **Year Restriction**: Cycles from previous years cannot be reopened

8. Credit Rating Rules

Credit Rating Assignment

- ✅ **Cycle-Specific**: Credit ratings are assigned per cycle
- ✅ **Tier Validation**: Credit rating must belong to a valid tier in the cycle's credit rating scheme
- ✅ **Borrowing Limit**: Borrowing limit is calculated as:
  - `Savings Balance × Credit Rating Multiplier`
  - Subject to cycle-specific borrowing limit policy maximum

Interest Rate Calculation

- ✅ **Tier-Based**: Interest rates are determined by credit rating tier
- ✅ **Term-Based**: Different interest rates apply to different loan terms
- ✅ **Range Validation**: Interest rates must fall within the tier's allowed range
- ✅ **Special Rule**: LOW RISK tier starts at 8% in new cycles

9. User Registration and Authentication Rules

Registration

- ✅ **Email Uniqueness**: Email addresses must be unique across all users
- ✅ **NRC Uniqueness**: National Registration Card (NRC) numbers must be unique (if provided)
- ✅ **Admin Approval**: New registrations require admin approval before access
- ✅ **Password Requirements**: Passwords are hashed using bcrypt

Authentication

- ✅ **Active Status Check**: Suspended members cannot log in
- ✅ **JWT Tokens**: Authentication uses JWT tokens with expiration
- ✅ **Role-Based Access**: Access to endpoints is controlled by user roles

10. File Upload Rules

Deposit Proof Upload

- ✅ **File Size Limit**: Deposit proof files must be within size limits
- ✅ **File Type Validation**: Only allowed file types can be uploaded (PDF, JPG, JPEG, PNG, GIF)
- ✅ **One Proof Per Declaration**: Each declaration can have one deposit proof
- ✅ **Storage**: Files are stored in `uploads/deposit_proofs/` directory
- ✅ **Status Workflow**:
  - Upload → `SUBMITTED` (declaration status remains `PENDING`)
  - Treasurer approval → `APPROVED` (declaration status changes to `APPROVED`)
  - Treasurer rejection → `REJECTED` (member can respond and resubmit)
- ✅ **Rejection Handling**: Rejected proofs can be approved after member provides response

11. Data Integrity Rules

Transaction Integrity

- ✅ **No Orphaned Records**: All journal entries must reference valid accounts

- ✅ **Reversal Tracking**: Reversed journal entries are tracked (not deleted)
- ✅ **Audit Trail**: All transactions include:
  - Creation timestamp
  - Creator user ID (if applicable)
  - Source type and reference
  - Cycle association

Referential Integrity

- ✅ **Foreign Key Constraints**: All foreign key relationships are enforced
- ✅ **Cascade Rules**: Appropriate cascade behaviors for related records
- ✅ **Unique Constraints**: Account codes, email addresses, and NRC numbers must be unique

12. Display and Calculation Rules

Balance Display

- ✅ **Social/Admin Fund**: Cards show accumulated payments (not outstanding balance)
- ✅ **Non-Negative Balances**: All balances are displayed as non-negative values
- ✅ **Currency Formatting**: All amounts are displayed in Kwacha (K) with proper formatting
- ✅ **Date Formatting**: Dates are parsed manually to avoid timezone issues (YYYY-MM-DD format)

Transaction History

- ✅ **Initial Requirements**: Marked with "Required Amount" badge (orange background)
- ✅ **Payments**: Marked with "Payment" badge (purple background)
- ✅ **Chronological Order**: Transactions displayed in reverse chronological order (newest first)
- ✅ **Source Tracking**: Each transaction shows its source type (declaration, deposit approval, etc.)
- ✅ **Penalty Display Rules**:
  - **Journal Lines**: All penalty-related journal entries are shown (charges and payments)
  - **PenaltyRecords**: Only `PENDING` and `APPROVED` penalties are shown as PenaltyRecord entries
  - **PAID Penalties**: Excluded from PenaltyRecord section (already represented by payment journal entries)
  - **Status Indicators**: Non-PAID penalties show status in description (e.g., "(APPROVED)", "(PENDING)")
- ✅ **Deduplication**: System prevents duplicate penalty entries in transaction history
- ✅ **Effective Month Display**: Effective months are displayed correctly (e.g., "January 2026") without timezone conversion issues

13. Late Submission Rules

Late Declaration Period

- ✅ **Declaration Deadline**: Declarations must be made by `monthly_end_day` (typically 20th) of the effective month
- ✅ **Late Penalty**: Declarations made after the deadline trigger automatic late declaration penalties (if `auto_apply_penalty` enabled)
- ✅ **Penalty Creation**: Late declaration penalties are created automatically when:

    - Declaration is submitted after `monthly_end_day`
    - `auto_apply_penalty` flag is enabled for the declaration phase
    - No duplicate penalty exists for the same member, penalty type, and effective month

Late Deposit Period

- ✅ **Deposit Window**: Deposits for a declaration's effective month are accepted from:
    - **Start**: `monthly_start_day` (typically 26th) of the effective month
    - **End**: `monthly_end_day` (typically 5th) of the **next** month
    - Example: For January 2026 declaration, deposit period is January 26, 2026 to February 5, 2026
- ✅ **Late Penalty**: Deposits submitted after the end date trigger automatic late deposit penalties (if `auto_apply_penalty` enabled)
- ✅ **Penalty Notes**: Late deposit penalties include period description: "Deposit period: {start_day}th of {effective_month} to {end_day}th of {next_month}"

Late Loan Application Period

- ✅ **Application Deadline**: Loan applications must be submitted by `monthly_end_day` (typically 20th) of the current month
- ✅ **Late Penalty**: Applications submitted after the deadline trigger automatic late loan application penalties (if `auto_apply_penalty` enabled)

Summary

The LUBOSS 95 system implements a comprehensive set of business rules to ensure:

1. **Financial Accuracy**: Double-entry bookkeeping with automatic balance validation
2. **Data Integrity**: Comprehensive validation at every step
3. **Audit Trail**: Complete transaction history with source tracking
4. **Flexibility**: Automatic account creation as needed
5. **Compliance**: Enforcement of village banking policies and procedures

All rules are enforced at the service layer, ensuring consistent behavior across all API endpoints and preventing invalid data from entering the system.

Recent Updates

Penalty System Enhancements (2026)

- **Status Workflow**: Implemented three-stage penalty status (PENDING → APPROVED → PAID)
- **Auto-Approval**: Cycle-defined penalties (Late Declaration, Late Deposits, Late Loan Application) are automatically created with APPROVED status
- **Transaction History**: PAID penalties are excluded from PenaltyRecord section to prevent duplicates (shown only as payment journal entries)
- **Duplicate Prevention**: Enhanced duplicate checking using member_id, penalty_type_id, date_issued, and notes

Late Deposit Period (2026)

- **Period Definition**: Deposits accepted from 26th of effective month to 5th of next month
- **Late Penalty Logic**: Penalties triggered only when deposit submitted after 5th of next month
- **Penalty Notes**: Include full period description for clarity

Deposit Approval Enhancements (2026)

- **Status Handling**: Both SUBMITTED and REJECTED proofs can be approved
- **Penalty Payment**: Automatic marking of APPROVED penalties as PAID when included in deposit
- **Journal Entries**: Corrected social/admin fund payment entries (credits reduce balance due)

Declaration Rules (2026)

- **Penalty Filtering**: Only APPROVED penalties included in declaration lists
- **Cycle-Defined Filter**: Respects `auto_apply_penalty` flag for cycle-defined penalties
- **Date Handling**: Fixed timezone issues in effective month display

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